Group 16: Social and Environmental Attributes that affect Tipping in New York

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Abstract—Tipping behavior in rideshare services is shaped by a range of social and environmental factors and plays a significant role in drivers' overall earnings. Despite its importance, the determinants of tipping behavior remain underexplored, particularly in relation to features such as rideshare, weather conditions, and service provider. This study aims to identify the key factors influencing tip amounts, with the goal of helping drivers better understand and potentially optimize their earnings. Leveraging publicly available rideshare and weather datasets from New York City, we investigate how tipping patterns vary by trip type (shared vs. single-rider), service provider (e.g., Uber, Lyft), and weather conditions (temperature, precipitation, wind speed, and rainfall). These datasets provide a rich set of features, including trip start and end times, fare and tip amounts, and daily weather metrics, allowing for a comprehensive multi-variable analysis. To analyze these relationships, we apply a combination of descriptive statistics, correlation analysis, and non-parametric statistical tests such as the Kruskal-Wallis H test and Mann-Whitney U test. Additionally, we employ predictive modeling techniques to assess the relative importance of each factor and to uncover patterns that may not be immediately visible through exploratory analysis alone. Data preprocessing involved cleaning, merging, and sampling steps to ensure the reliability and usability of the data. Our findings provide meaningful insights into the dynamics of tipping behavior in urban rideshare contexts and highlight the role of both environmental and social variables in shaping financial outcomes for drivers. By understanding the conditions under which tips are more likely to be higher or lower, both rideshare companies and drivers can make more informed decisions.

Index Terms—Group 16, Rideshare Data, Tipping Behavior, Social Factors, Environmental Factors, Weather Impact, Predictive Modeling, Deep Learning, Data Analysis, Machine Learning, Statistical Testing, PyTorch, Neural Networks, Urban Mobility.

1 Introduction

Tipping behaviour in the rideshare industry plays a significant role in drivers' earnings, yet the factors influencing tip amounts remain insufficiently explored. Our objective is to determine what parameters are important for predicting tip amount, so drivers can attempt to optimize these parameters. We will use data from past taxi trips in NYC including tip amount, as well as weather data for NYC to predict tip amount. From there, we will analyze the model to determine which parameters are most important, and which parameters have a high correlation to tip amount. We will consider our project successful if we can accurately determine which parameters are important for predicting tip amount, and provide a strategy for drivers to maximize their tips.

The goal of our research is to identify the key factors that influence tipping behavior in taxi services and determine how drivers can optimize their earnings from tips. In specific, we aim to explore which variables within the control of taxi drivers are they able to exploit to maximize the tips they receive. For example, if there is precipitation, would it be a good, bad, or neutral time to offer your services? What app (Uber, Lyft, etc) is most profitable? And so on.

Other existing work has also explored taxi data in NYC, including how to maximize tips, but many only focus on yellow-taxi data, rather than other services like Uber and Lyft. Furthermore, there are still many features that haven't been considered when trying to understand how to maximize tips. For example, weather isn't a widely explored parameter. Our novelty will stem from a combination of using a different dataset that focuses on for-hire services (Uber, Lyft, etc), as well as focusing on these unexplored features for how to maximize tips.

To summarize, we plan to answer the following research questions (RQs):

- **RQ1:** How does the tip amount differ between rideshare (shared rides) and single-rider trips?
- **RQ2:** Is the tip amount dependent on the service provider (Uber, Lyft, etc)?
- **RQ3:** How does rain impact tipping behaviour?

To answer these questions, we performed statistical tests, correlation analysis, and extracted feature importance from a trained neural network designed to predict tip percentage. statistical tests were first used to identify if the RQ-related features had a signficiant impact on the tip percentage. Correlation analysis was then used to identify any direct correlation between RQ-related features and the tip percentage. Finally, to investigate indirect correlations, we trained a neural network to predict tip amount and then extracted feature importance from it.

2 Related Work

Understanding tipping behavior in taxi services has been the focus of multiple research efforts, each contributing unique insights relevant to our study.

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[3] investigated preferences between shared and non-shared ride-hailing services, exploring how trip costs, travel times, and socio-economic factors influence user choices. Their novel methodology linked individual characteristics from household travel surveys with ride data, avoiding reliance on preference surveys. This approach aligns with our study's aim to analyze how external factors, like trip types, influence tipping behavior. Their findings on user preferences can provide critical insights into factors impacting tipping behavior in shared versus single rides.

[4] examined how weather conditions affect taxi tipping behavior in New York City. They utilized a cumulative logit model to categorize tipping behavior and found that adverse weather, such as rain or snow, impacted tip amounts. Their methodology and insights on how environmental factors influence tipping behavior are highly relevant to our research question regarding weather's effect on tipping. However, the datasets used, pre-processing, and analysis techniques differ from those employed in our study.

[5] analyzed the impact of trip characteristics and weather conditions on the ride-sourcing network performance of Uber and Lyft in Philadelphia. Using statistical and network analysis techniques, the study provided empirical evidence on how external factors influence demand and supply dynamics. This paper's integration of trip and weather data offers a methodological framework that supports our analysis of tipping behaviors influenced by environmental variables. Additionally, it emphasizes the need for integrating multiple data sources, a strategy we also employ in our study.

[1] explored psychological and contextual factors influencing individuals' decisions to participate in ridesharing. Their use of structural equation modeling to assess factors like economic benefits, transportation anxiety, and trust provides foundational insights into user motivations. This research complements our study by offering a baseline understanding of user behavior in rideshare services, which can enhance our interpretation of tipping trends and behavior patterns in our analysis.

In summary, these studies offer valuable methodological insights and contextual understanding that enrich our analysis of social and environmental attributes affecting tipping behavior in NYC rideshare services.

3 Methodology

3.1 Data Pre-Processing:

After collecting the TLC Trip Record dataset and NYC weather dataset, we performed a series of preprocessing steps:

- **Data Cleaning:** Removed rows with missing or clearly incorrect values (e.g., negative fares or tip amounts), as they're MCAR.
- **Datetime Formatting:** Converted all datetime fields to a standard format and extracted additional features like hour of the day, day of week, and month.
- **Service Identification:** Mapped the license type to the service providers (e.g., Uber, Lyft).
- **Data Merging:** Joined weather data (temperature, precipitation, wind speed) with trip records based on the TLC dataset.
- **Random Sampling:** We randomly sampled roughly 5,000,000 rows from the cleaned and merged dataset for analysis and modeling, due to a very large dataset, to allow for faster computational times.

This preprocessing ensured data consistency and enabled a meaningful analysis of tipping behavior across different time frames and environmental conditions.

3.2 Data Exploration

Upon pre-processing, we conducted some data exploration on our processed dataset, to better understand the distribution and behavior of key features. Histograms, boxplots, and time-based aggregations were used to identify outliers and understand the trends in tipping patterns across services and environmental conditions. Categorical variables such as service type and ride-sharing flags were converted using one-hot encoding, while numerical features were normalized where necessary for modeling.

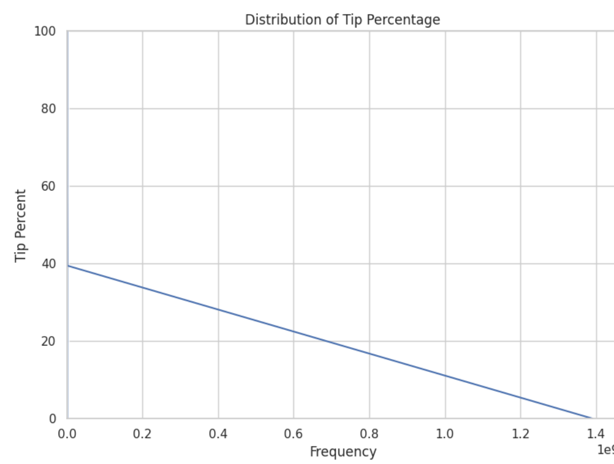

Fig. 1. Distribution of Tip Percentage

3.3 Correlation and Statistical Analysis

Correlation

We created a correlation matrix to analyze the relationships between tipping behavior and key features including weather metrics, service providers, and ride-types. A correlation heatmap was generated to visually assess the strength and direction of relationships between these variables. This allowed us to identify notable patterns between predictors.

Hypothesis Testing

We conducted statistical hypothesis testing using the Mann-Whitney U test and Kruskal-Wallis H test to validate the significance of our RQ related features. These tests helped determine whether tipping behavior is meaningfully impacted by features related to our RQs statistically.

The Mann-Whitney U test was used for binary comparisons, while the Kruskal-Wallis H was used for multi-group comparisons.

Each test examined a specific research question using a significance threshold of $\alpha = 0.05$. The results showed whether to reject or fail to reject the null hypothesis. This provided a statistical foundation for our findings and confirmed whether tipping behavior differs significantly based on ride-sharing, service provider, and weather conditions.

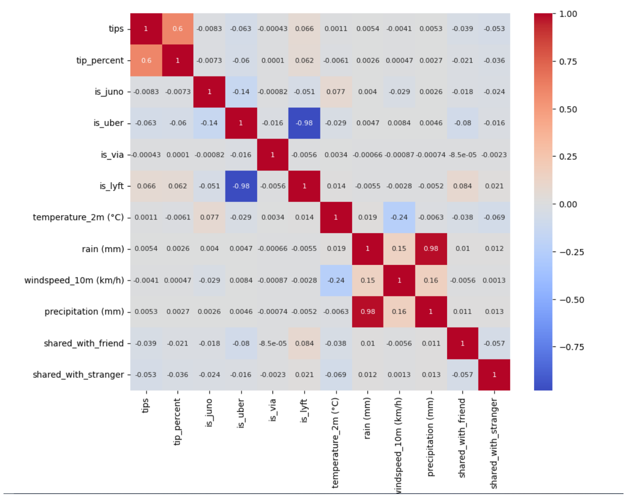

Fig. 2. Correlation matrix of tipping-related variables

TABLE 1
Statistical Test Results on Tipping Behavior

Research Question	Test Used	p-value	Sig.?
1: Shared vs. Non-Shared	Mann-Whitney U	0.000	Yes
2: Service Provider Effect	Kruskal-Wallis	0.000	Yes
3: Rain vs. No Rain	Mann-Whitney U	2.962e-10	Yes

3.4 Predictive Model

We developed and trained a deep learning model using PyTorch to better understand the importance of different variables and predict tipping behavior. We selected a neural network for this task due to its ability to capture complex, nonlinear relationships within the data. Although we considered simpler models such as linear regression or tree-based models during our initial experimentation, the deep learning approach demonstrated better flexibility and learning capacity for the multidimensional features in our dataset.

Model

We implemented a feedforward neural network consisting of five fully connected layers with ReLU activation functions. The network was designed to model complex, nonlinear relationships between features and the target variable. The primary prediction target was the tip percent, which is a normalization of the tip amount based on base fare, tolls, and taxes.

Model Training

- **Data Preprocessing:** The dataset was cleaned and transformed to include only numeric and relevant features. One-hot encoding was applied to categorical variables like service provider.
- **Train-Test Split:** The data was split into 80% training and 20% testing sets using stratified sampling to preserve distribution.
- **Training Configuration:** The model was trained using the Adam optimizer and Mean Squared Error (MSE) loss function over 100 epochs, with a batch size of 128.
- **Evaluation Metrics:** Performance was evaluated using RMSE (Root Mean Squared Error) to measure prediction error (since it is intuitive to humans), and R^2 score to assess explained variance.

The neural network demonstrated good generalization capabilities, learning complex relationships between environmental, temporal, and social ride-sharing features and tip behavior. It demonstrated the capacity to generalize well across a range of input conditions, offering competitive performance relative to other models.

3.5 Model Extraction

To interpret our neural network model and understand which features most significantly influenced tipping behavior, we used two main approaches: weight-based analysis and SHAP (SHapley Additive Explanations).

Weight-Based Importance: We analyzed the absolute values of the weights in the first fully connected layer of the neural network. Features with larger cumulative absolute weights were considered more influential. This provided a coarse approximation of how input variables initially contribute to the learned patterns within the model. Through this, we generated a graph showing the most influential factors in our prediction model.

SHAP Value Analysis: We used SHAP (DeepExplainer) to compute the marginal contribution of each feature to individual predictions, in order to gain a deeper, model-aware interpretability. This method attributes changes in the output (predicted tip percentage) to each input feature, offering both local and global explanations.

We generated:

- **MSE Error Rate:** We evaluated the model using Mean Squared Error (MSE), quantifying the difference between predicted vs actual tip percetages.
- **Waterfall Plots:** We visualized the contribution of individual features to a single prediction, showing how each input pushes the prediction higher or lower from the base (expected) value.

The combination of predictive modeling and visualizations helped us to derive meaningful insights regarding both model performance and the underlying dataset. These tools allowed us to assess the accuracy of our predictions, identify the most influential features, and uncover patterns in tipping behavior across different social and environmental contexts.

4 Dataset

We used two primary datasets to answer our research questions.

Our first dataset is the TLC Trip Record Data, which contains all rideshare taxi data. This relatively large dataset includes numerous attributes and categories that allow us to answer our research questions accurately. For example, it provides data on tip amount and service provider (Uber, Lyft, etc.). The dataset offers insights into various trip characteristics, such as whether the trip was a shared or single-rider ride, which is crucial for exploring the correlation between trip types and tipping behavior. [2] It is a publicly available dataset collected annually by the New York City government .

The second dataset consists of weather data from New York City, which includes information on temperature, precipitation, wind speed, and other weather conditions. This dataset is used to address our third research question: how weather affects tipping practices in New York. By combining this dataset with the TLC Trip Record Data, we aim to explore any correlations between weather conditions and tipping behavior. [6] The dataset covers weather data for New York City from 2016 to October 25, 2022.

Data Pre-Processing

We performed several pre-processing steps to ensure data quality and compatibility. To ensure that the datasets were in a suitable format for analysis, several pre-processing steps were undertaken, which included data cleaning, handling missing values, and merging the two datasets.

Firstly, for the weather data, we removed rows with missing values and duplicates, converted the time column to datetime format, and retained one unique entry per day, eliminating duplicate rows. We then saved the data as a new CSV file for use in further processing.

Next, for our TLC data, we followed the same format as prior, removing rows with missing values and duplicates, converting the time column to datetime format, and retaining one unique entry per day, eliminating duplicate rows. Further, we also had to deal with converting license types to services (Uber, Lyft etc..)

After cleaning both datasets, we merged them on the date field to create a unified dataset, enabling us to explore the relationship between weather conditions and tipping behaviour more effectively.

We also conducted some basic statistical analysis with a focus on RQ-related features:

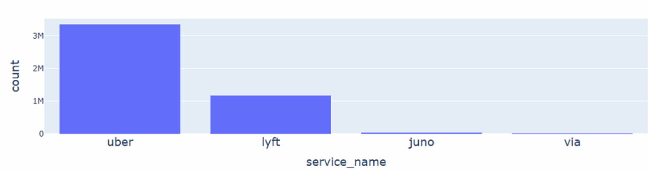

Fig. 3. Number of taxi rides by service provider

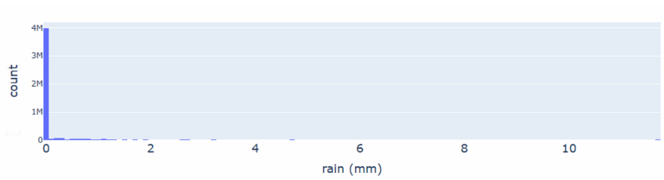

Fig. 4. Number of taxi rides by rain

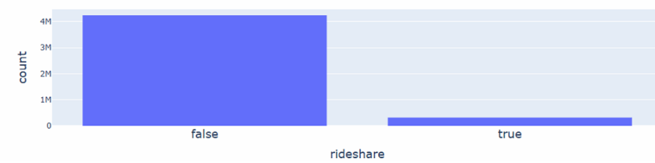

Fig. 5. Number of taxi rides by rain

After pre-processing and merging, a random sample of 5,000,000 rows was taken from the merged dataset. This sample size was chosen to provide a representative subset of the data while reducing computational load for further analysis.

5 Experiments and Results

Experiment Set-Up

Our experiment was conducted using a cleaned and preprocessed dataset of 5,000,000 randomly sampled rows from NYC taxi rides and weather data. We used the tip percentage as the prediction target. The feature set included service provider indicators, weather metrics (precipitation, temperature, wind speed), and ride-sharing flags.

We trained a feedforward neural network using PyTorch, consisting of five fully connected layers with ReLU activations. The model was trained for 100 epochs using the Adam optimizer and Mean Squared Error (MSE) loss function. An 80/20 train-test split was applied with a batch size of 128.

Performance was evaluated using:

- **RMSE (Root Mean Squared Error)** – quantifying average prediction error.
- **R^2 Score** – representing how much variance is explained by the model.
- **MSE (Mean Squared Error)** – for deeper model diagnostics.

Model Performance

Our neural network achieved the following performance metrics on the test set:

- RMSE: 0.0153
- R^2 Score: 0.9745
- MSE: 0.000233

These results suggest that the model is able to explain approximately 97% of the variance in tip percentage.

To further interpret our model, we analyzed the absolute values of the weights in the first layer of the neural network. This provided an additional perspective on the features that the model emphasized during the training. The weight-based importance helped identify which input variables had the most influence on the model's internal representations. The results aligned well with the top features identified in our previous correlation analysis, offering validation for our initial assumptions. This approach also allowed us to better understand how each feature impacted tip percentages, serving as a useful cross-check tool for previous experiments.

These observations not only validated our hypotheses but also added transparency to the model's decision-making process, helping to identify which characteristics have the greatest impact on tipping behavior.

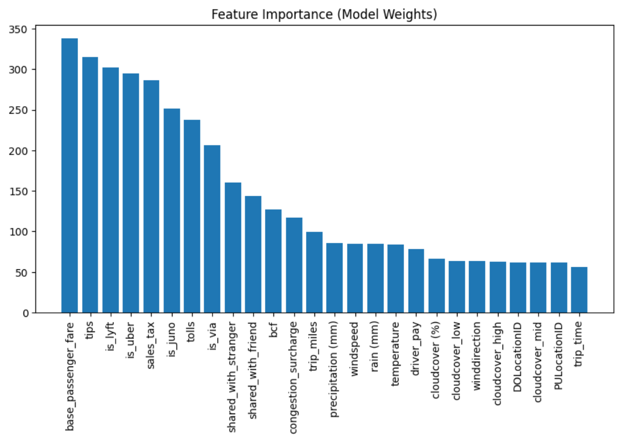

Fig. 6. Feature importance plot based on the absolute weights from the neural network layer.

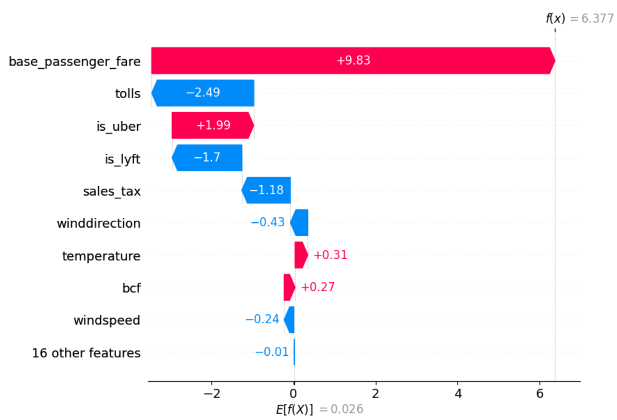

Fig. 7. SHAP waterfall plot showing the contribution of each feature across 5000 random samples.

Result by Research Question

RQ1: Do shared and non-shared rides differ significantly in tipping behaviour?

Using the Mann-Whitney U test, we found a statistically significant difference in tip percentages between shared and non-shared rides ($p = 0.000$). Boxplot visualizations also showed that non-shared rides tend to receive higher tips on average.

Alongside, the correlational analysis supports our findings showing the correlation between tips and shared with a stranger is -0.053, and shared with a friend is -0.039. Both of these relations are negative, which suggest that sharing a ride —particularly with strangers— may be associated with slightly reduced tipping behaviour.

Furthermore, from our feature importance analysis, we can see that both shared with a stranger, and shared with a friend is amongst the top features influencing tip percentage. The combination of these results help answer our research question that yes, tipping behaviour does significantly differ between shared and non-shared rides, showing that non-shared rides are more profitable for drivers in terms of tipping.

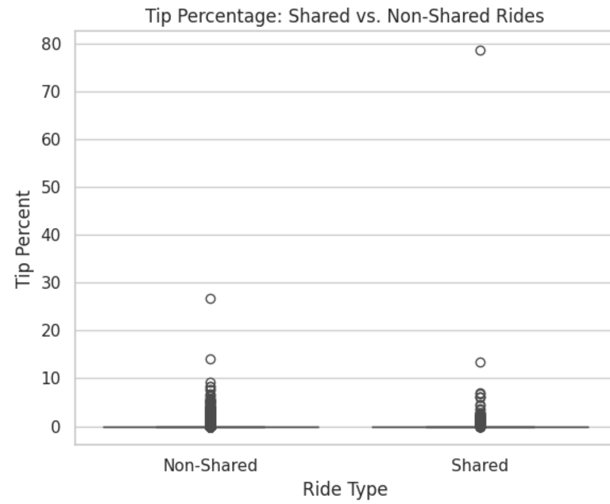

Fig. 8. Tip Percentage: Shared vs. Non-Shared Rides

RQ2: Are tip amounts affected by the service provider?

A Kruskal-Wallis H test revealed significant differences in tipping patterns across Uber, Lyft, Juno, and Via ($p = 0.000$). Our model indicated that Uber trips generally yielded higher tipping behaviour than Lyft. However, correlation analysis showed a weak relationship between service provider and tipping: Uber (-0.063), Lyft (0.066), Juno (-0.0083), and Via (-0.0043).

Looking at our feature importance graph , we can see that `is_uber` and `is_lift` are amongst the one of the most important features.

Alongside, looking at the SHAP waterfall graph, we can see that the difference between the service providers especially Lyft and Uber, effect the tip percentages. These results show that yes, tip amounts are affected by service provider, showing that Uber is the platform with the highest tip percentage. While this does disagree with the correlation analysis, we believe that the SHAP waterfall graph yields a more reliable conclusion, since it can handle more complex relationships, and the correlation analysis yielded a relatively weak correlation.

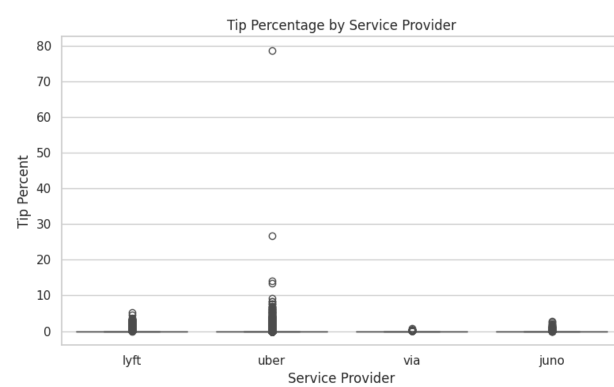

Fig. 9. Boxplot of Tip Percentage by Service Provider

RQ3: How does rain impact tipping behaviour?

We conducted a Mann-Whitney U test, and it showed a statistically significant difference in tipping between rainy and non-rainy rides. We found the p values to be p = 2.962e-10 when using the tip percentage and 8.374e-12 when using the tip dollar amount.

There was no significant direct correlation between tipping and rain found in the correlation analysis.

In the feature importance graph, weather variables are near the middle in terms of importance in the model. Therefore, for RQ3, the answer is: rain does have a statistically detectable impact on tipping, but the effect is not great. More specifically, people tend to tip less on rainy days.

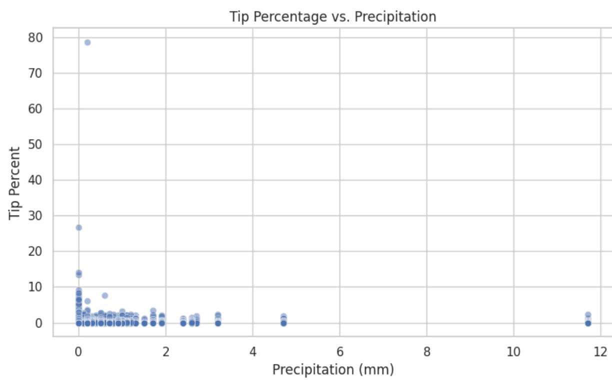

Fig. 10. Tip Percentage vs. Precipitation

Threats to the Validity

Although we designed and executed our experiments with care to ensure reliability, certain limitations remain, and we acknowledge several potential threats to the validity of this study, which may influence the interpretation or generalization of our findings.

- **Sampling bias:** While our dataset was randomly sampled, there may still be under-representation of certain trip types or weather events.
- **Temporal granularity:** Aggregating weather data at the daily level may dilute its true impact on trip-specific tipping. For example, if it rained in the morning, a taxi ride in the afternoon may still be labelled as occurring during a rainy day.
- **Feature leakage:** Variables like total tip amount are correlated with other pricing fields; we took care to normalize tips but note this dependency.
- **Model generalizability:** While the model performed well on our dataset, results may not generalize to other cities or future periods due to shifts in user behavior or platform policies.

6 Group Member Contributions

Adithya has handled data collection and preprocessing, including the collection and cleaning of raw data. He ensured the data was correct for analysis and addressed any mishandling or outliers. Andrew has conducted additional data analysis, including basic statistics, statistical tests, and correlational analyses. He has investigated relationships among variables and their influence on the model, ensuring that the analysis provides deeper insights into the data and allows us to make predictions. He also edited the final report and the final presentation. Andrew was also responsible for building and training the prediction model, testing its performance, and fine-tuning it to achieve optimal results. Raghab used the model outputs to extract meaningful insights and identify key parameters. He also wrote the draft of the final report.

7 Replication Package

Link to github repo: https://github.com/Adithya-Raj0604/Tipping-in-New-York

8 Conclusion and Future Work

Conclusion

In our study, we explored the social and environmental factors that influence tipping behaviour in New York City's rideshare services. We were able to examine how variables such as ride type (shared vs. non-shared), service provider (Uber, Lyft, etc.), and weather conditions (rain, temperature, wind speed) impact tip percentages, by integrating and analyzing large-scale rideshare data, alongside a weather dataset.

Our results show that non-shared rides receive statistically higher tips than shared rides, with a Mann-Whitney U test confirming this difference as significant.

Furthermore, the service provider plays a significant role in tipping behaviour, with Uber trips typically receiving higher tips compared to other platforms like Lyft and Juno.

Weather conditions, in particular precipitation, was also found to have a statistically significant, effect on tipping, with rainy conditions slightly reducing tip percentages.

Our predictive model, a neural network trained using PyTorch, achieved high accuracy with an R^2 score of 0.9745 and an RMSE of 0.0153, showing that it has a strong predictive capabilities. The use of SHAP values and weight-based feature importance provided further transparency into the model's decision-making, confirming that variables such as service provider, ride type, and weather metrics are among the most influential features.

Overall, our research contributes a novel perspective on tipping behaviour in urban rideshare services, combining environmental and social factors with machine learning to provide actionable insights for drivers and companies

Future Work

While our findings offer useful insights for both drivers, several limitations remain, such as, the dataset was sampled to 5,000,000 rows for efficiency, which may exclude edge cases or underrepresented scenarios.

In order to further improve our study for the future, we can implement a multitude of steps:

- **Granular Weather Data:** Integrating hourly weather data may provide more precise insights into how immediate weather conditions affect tipping.
- **Geospatial Analysis:** Adding spatial features such as pickup/drop-off location or neighborhood income levels could uncover new socio-economic patterns.

- **Temporal Patterns:** Future models could examine how tipping behavior evolves over time (e.g., seasonal trends or holidays).
- **Model Benchmarking:** Comparing neural networks with other advanced models like XGBoost or ensemble methods may reveal more efficient or interpretable models.
- **Policy Simulation:** Developing tools or simulations that help drivers make real-time decisions (e.g., when and where to drive) based on expected tip behavior could turn insights into actionable strategy.

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